

BAGDARIN, Russian FederationWMO: **305540**Lat: **54.442N**Lon: **113.581E**Elev: **902**StdP: **90.94**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.8	-39.6	-44.0	0.1	-41.5	-42.0	0.1	-39.4	5.2	-17.8	4.3	-18.4	0.0	0	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	27.2	16.4	25.4	15.6	23.7	14.9	18.1	23.8	17.0	22.8	16.1	21.7	2.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	12.8	19.9	15.0	11.9	19.1	13.9	11.1	18.4	55.1	24.0	51.6	22.6	48.5	21.7	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.2	4.4	-44.3	30.2	2.4	2.1	-46.0	31.7	-47.4	33.0	-48.7	34.1	-50.5	35.7		
(5)			-44.2	20.2	2.1	1.5	-45.7	21.3	-46.9	22.2	-48.0	23.0	-49.5	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.1	-27.6	-22.3	-13.7	-2.8	5.6	12.9	15.8	13.1	5.4	-4.5	-18.0	-26.5		
	DBStd	16.13	5.28	5.33	6.63	5.08	3.99	3.53	2.66	3.08	4.23	5.45	6.87	5.72		
	HDD10.0	5920	1166	903	736	383	144	14	1	9	147	449	838	1131		
	HDD18.3	8567	1424	1137	994	633	394	166	84	164	387	707	1088	1389		
	CDD10.0	403	0	0	0	0	8	99	182	104	9	0	0	0		
	CDD18.3	8	0	0	0	0	0	1	7	1	0	0	0	0		
	CDH23.3	436	0	0	0	0	7	127	237	61	3	0	0	0		
	CDH26.7	72	0	0	0	0	1	18	47	7	0	0	0	0		
(14) Wind	WSAvg	1.3	0.5	0.9	1.5	2.1	2.3	1.6	1.4	1.2	1.5	1.3	0.8	0.4		
(15) Precipitation	PrecAvg	396	3	2	3	16	34	78	103	100	37	10	7	3		
	PrecMax	562	8	8	11	49	90	192	201	207	102	30	30	12		
	PrecMin	192	0	0	0	1	3	8	31	15	5	0	0	0		
	PrecStd	88	2	2	3	12	25	49	49	49	20	8	5	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.5	-2.3	8.1	16.3	24.4	28.9	30.1	27.4	23.1	13.5	3.7	-7.4		
		MCWB	-11.4	-5.6	1.7	6.5	12.3	16.5	18.2	16.2	12.3	6.2	0.0	-9.3		
	2%	DB	-13.9	-6.1	4.3	12.2	21.1	26.5	27.9	25.1	19.8	9.7	-1.2	-11.5		
		MCWB	-15.0	-8.4	-0.7	4.3	10.5	15.3	17.1	15.5	11.0	3.6	-3.9	-12.7		
	5%	DB	-16.6	-9.1	1.1	9.6	18.5	24.6	26.0	23.0	16.9	7.0	-4.5	-14.8		
		MCWB	-17.4	-11.0	-3.0	2.8	9.0	14.4	16.5	14.8	9.7	2.0	-6.3	-15.7		
	10%	DB	-19.0	-11.8	-1.7	7.1	15.9	22.7	24.2	21.1	14.4	4.4	-7.0	-17.9		
		MCWB	-19.5	-13.1	-4.9	1.3	7.7	13.6	15.9	14.3	8.6	0.3	-8.5	-18.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.4	-5.6	1.9	6.9	12.7	18.8	19.9	18.3	14.2	6.3	-0.1	-9.1		
		MCDB	-9.6	-2.6	7.7	15.6	23.1	26.0	26.0	22.4	21.2	12.9	3.6	-7.5		
	2%	WB	-15.0	-8.4	-0.7	4.7	11.0	16.9	18.6	17.0	12.2	3.9	-3.8	-12.7		
		MCDB	-14.0	-6.2	4.0	11.6	19.8	23.3	25.2	22.7	17.2	9.3	-1.7	-11.4		
	5%	WB	-17.4	-10.9	-2.9	3.1	9.5	15.7	17.6	16.0	10.6	2.2	-6.2	-15.6		
		MCDB	-16.6	-9.2	0.9	9.2	17.4	21.7	23.4	21.2	15.8	6.7	-4.7	-14.9		
	10%	WB	-19.6	-13.2	-4.8	1.5	8.1	14.5	16.7	15.0	9.0	0.4	-8.5	-18.4		
		MCDB	-19.0	-11.8	-1.7	6.8	15.3	21.0	22.3	19.6	14.1	4.0	-7.1	-17.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	19.7	19.3	15.2	15.0	15.8	14.6	14.3	14.6	14.6	14.6	12.8		
		MCDBR	15.4	21.2	19.9	19.2	21.5	20.8	19.3	18.8	19.2	17.1	15.6	14.3		
		MCWBR	14.4	18.8	15.2	12.0	11.8	10.6	9.8	10.0	10.9	11.3	12.9	13.1		
		MCDBR	15.3	20.8	19.4	18.4	19.7	16.6	15.5	15.9	16.7	16.3	15.1	14.2		
	5% WB	MCWBR	14.3	18.5	14.8	11.7	11.1	9.2	8.8	9.2	9.7	10.9	12.5	13.0		
(40) Clear-Sky Solar Irradiance	taub		0.235	0.265	0.294	0.371	0.459	0.461	0.405	0.375	0.323	0.280	0.253	0.230		
	taud		2.177	2.159	2.166	2.109	1.960	2.072	2.306	2.371	2.463	2.448	2.249	2.131		
	Ebn at Noon		730	825	881	848	785	790	835	841	850	815	697	640		
	Edh at Noon		60	88	111	137	169	153	119	105	84	66	56	50		
(44) All-Sky Solar Radiation	RadAvg		1.04	2.06	3.66	5.14	5.53	5.74	5.31	4.51	3.40	2.23	1.21	0.77		
	RadStd		0.05	0.06	0.10	0.18	0.35	0.37	0.48	0.32	0.25	0.12	0.06	0.04		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-3

Nomenclature: See separate page